AI Powered Mental Health Therapist Chatbot using Generative AI

Team Members

1. F. Anil Kumar – 22N31A7313
2. R. Varaprasad – 22N31A7356
3. S. Srujana – 23N35A7306

IV Year B.Tech-I Semester – B. TECH (AIML)

Under the Guidance of
Dr. A. Nagaraju
H.O.D

Agenda

1. Abstract
2. Introduction
3. Existing System
4. Drawbacks of Existing System
5. Proposed System
6. Advantages of Proposed System
7. Requirements Specifications
8. UML Diagrams

Abstract

Mental health awareness and timely self-care are increasingly vital in today's fast-paced world, yet access to professional mental health support remains limited for many. To address this gap, we present an AI-powered Mental Health Chatbot designed to assist users in identifying symptoms, answering mental wellness queries, and guiding them toward self-care resources in a safe, private, and accessible way. This chatbot acts as a digital mental health companion, fine-tuned to respond empathetically and informatively. It interprets user queries in real-time, performs sentiment analysis, and provides tailored responses along with self-care suggestions and mental health tips. The interface is built using React.js and Bootstrap, ensuring a responsive and calming user experience, while the Django backend manages authentication, data flow, and AI integration seamlessly. This solution aims to provide accessible mental health support and bridge the gap for those with limited access to professional help.

Introduction

Mental health has become a major concern in the modern world, with increasing numbers of individuals experiencing stress, anxiety, and emotional challenges. Despite growing awareness, access to timely and professional mental health support is still limited for many due to social, economic, or geographical barriers. Technology offers an opportunity to address this gap through innovative, scalable solutions. This project introduces an AI-powered Mental Health Chatbot designed to provide immediate, accessible, and empathetic mental health support. The chatbot assists users in recognizing symptoms, answering wellness-related queries, and suggesting self-care practices. It uses natural language processing and sentiment analysis to interpret emotions and respond appropriately. The goal is to create a safe and private space for users to engage with mental health resources. Built with React.js and Bootstrap, the interface is user-friendly and calming. A Django backend manages data flow, authentication, and AI services efficiently. This solution aims to complement traditional support systems and make mental health care more inclusive and reachable.

Existing System

- **Limited Access to Professionals**
  Many individuals face difficulties accessing therapists or counselors due to high costs, long wait times, or geographic limitations.
- **Stigma Around Seeking Help**
  Social stigma often prevents people from openly discussing mental health issues or seeking professional support.
- **Lack of Immediate Support**
  Traditional mental health services may not be available 24/7, leaving users without help during critical moments.
- **Manual Symptom Assessment**
  Users typically need to self-assess or fill out static questionnaires, which may not provide real-time or adaptive feedback.
- **Minimal Use of Technology**
  While some apps and hotlines exist, many lack AI capabilities, personalization, or emotional intelligence in their responses.

Drawbacks of Existing System

•**Limited Accessibility**
Professional mental health services are often costly, have long waiting times, and are unavailable in many rural or underserved areas.

•**Stigma and Social Barriers**
Many individuals avoid seeking help due to fear of judgment, social stigma, or cultural taboos surrounding mental health.

•**Lack of Immediate Support**
Traditional services typically operate during specific hours, leaving users without help during emotional crises or off-hours.

Proposed System

- **24/7 Accessible Support**
  Users can access the chatbot anytime for immediate guidance, without waiting for appointments or availability.
- **Emotionally Aware Interaction**
  Real-time sentiment analysis allows the chatbot to understand user emotions and respond empathetically.
- **Personalized Self-Care Suggestions**
  Based on user input, the chatbot offers tailored mental wellness tips, coping strategies, and self-care activities.
- **Safe and Private Environment**
  The system ensures user privacy, encouraging open and honest conversations without fear of judgment.
- **User-Friendly Interface**
  Developed using React.js and Bootstrap with Django backend, the system delivers a smooth, calming, and secure user experience.

Advantages of Proposed System

- **24/7 Availability**
Provides round-the-clock mental health support, allowing users to seek help anytime, especially during emotional emergencies.

- **Personalized and Empathetic Responses**
Uses sentiment analysis and natural language processing to understand user emotions and deliver tailored, compassionate replies.

- **Accessible and Cost-Effective**
Offers a free or low-cost alternative to traditional therapy, making mental health support more inclusive and reachable.

System Requirements

Software Requirements

Frontend:
- React.js (Version 18.x)
- Bootstrap (Version 3.x)
- Node.js (Version 16.x or later)
- npm

Backend:
- Python (Version 3.9 or later)
- Django REST Framework (Version 3.14.x)

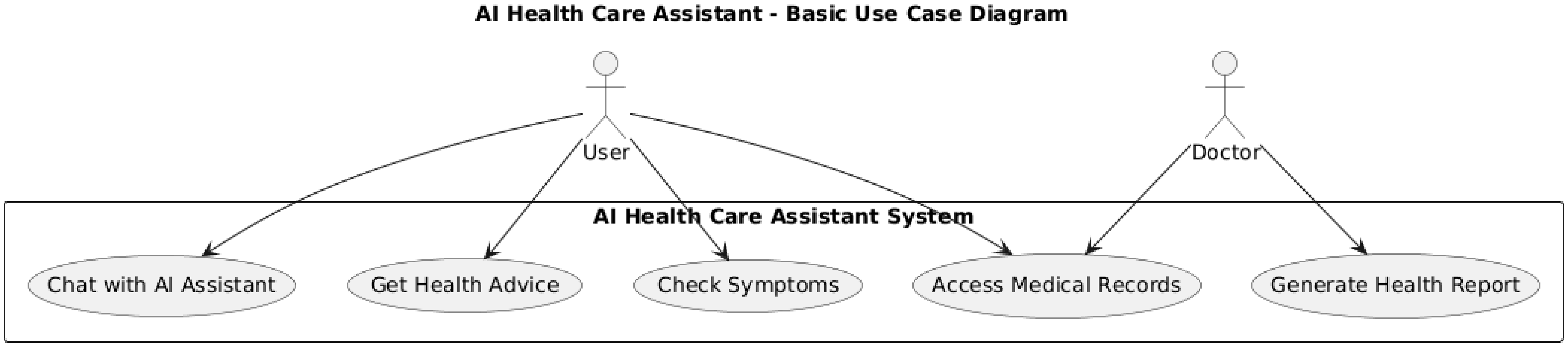
AI and NLP:
- Transformers library , Hugging Face Transformers.

Hardware Requirement Specifications (HRS)

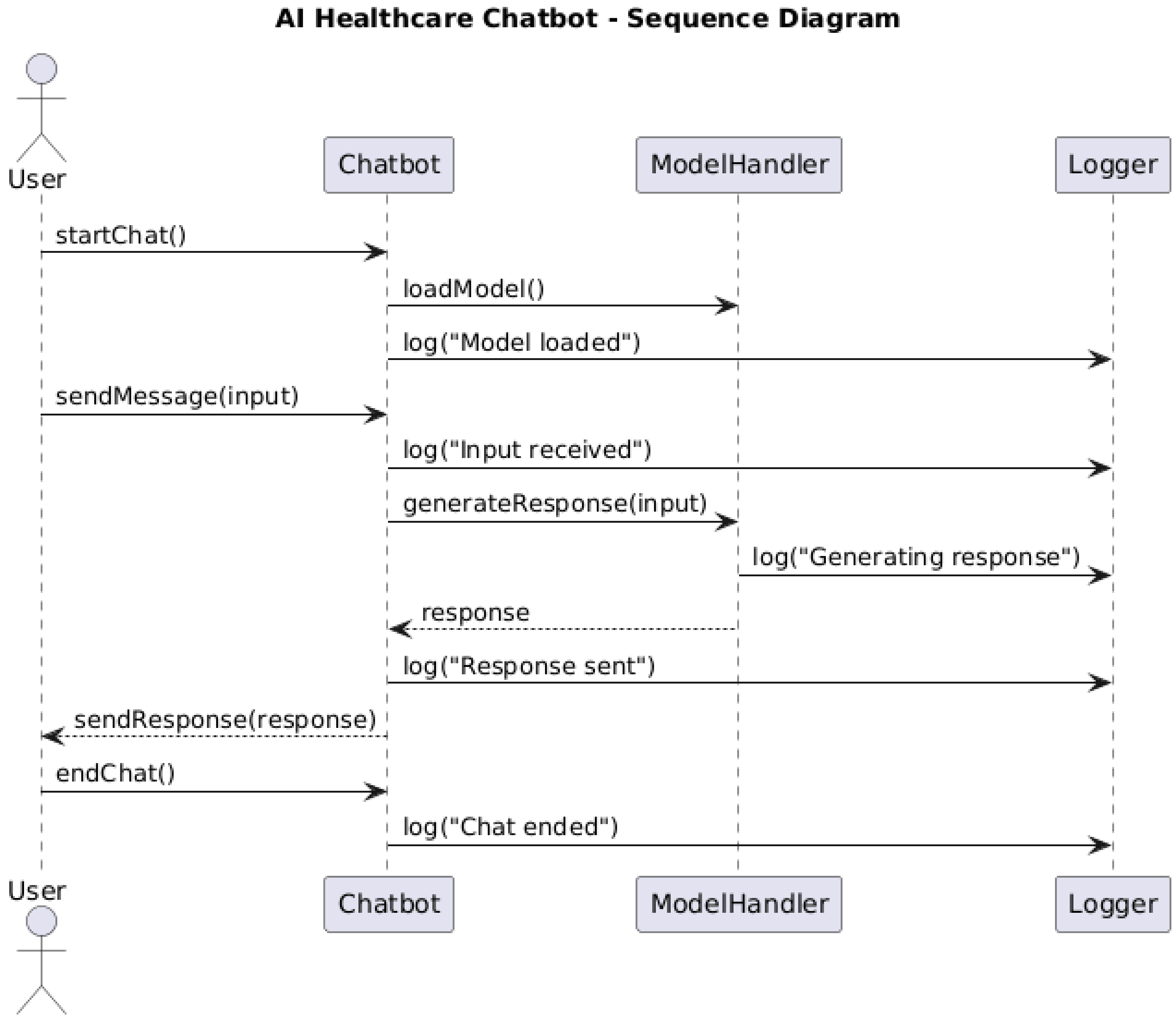
- Processor: Intel Core i5
- RAM: Minimum 8 GB
- Storage: 100 GB SSD
- GPU (AI model training/fine-tuning): NVIDIA GTX 1060 or higher

UML DIAGRAMS

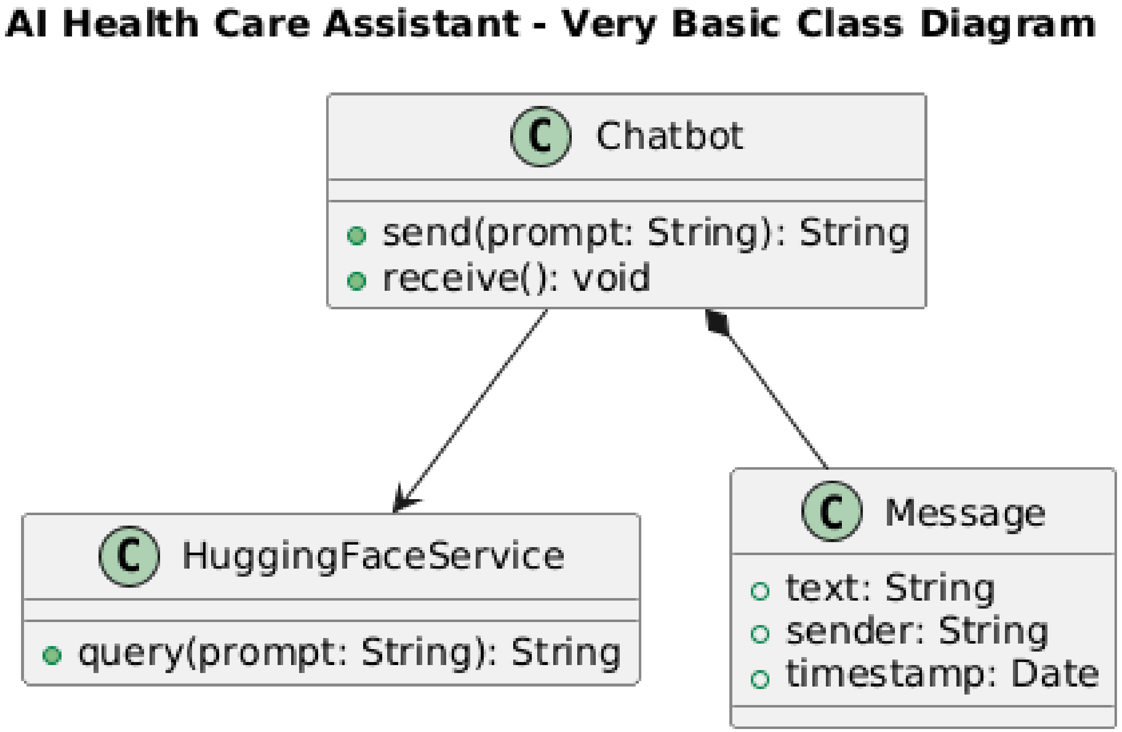
Use Case Diagram:



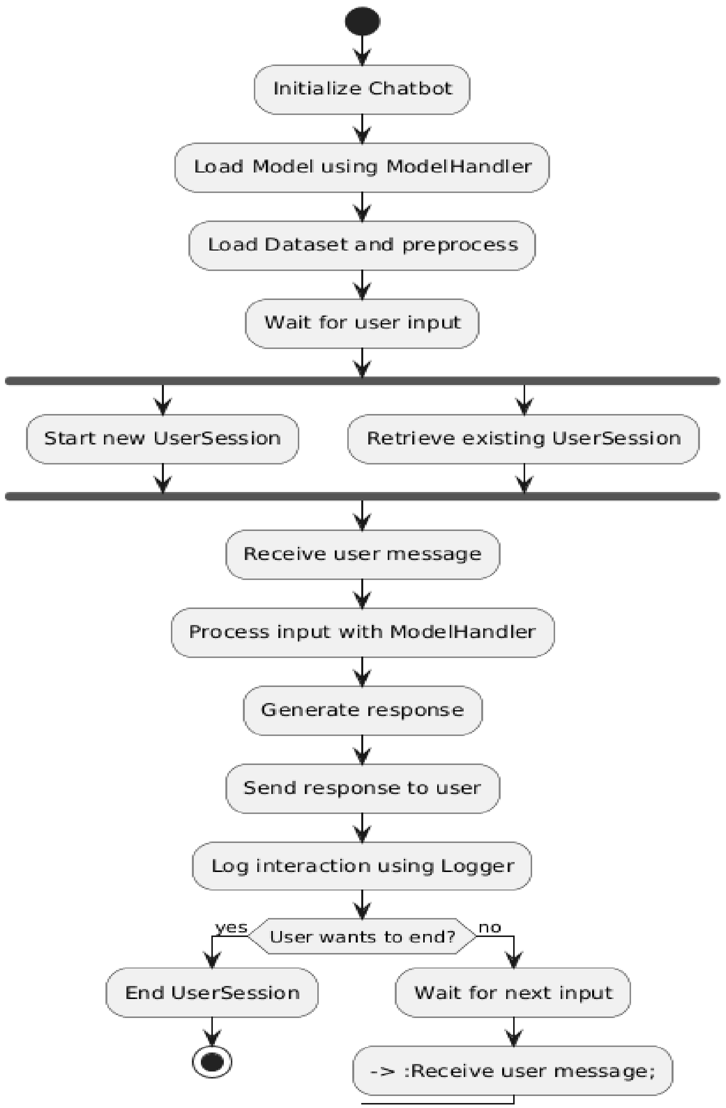
Sequence:


AI Healthcare Chatbot - Sequence Diagram
User
Chatbot
ModelHandler
Logger
startChat()
loadModel()
log("Model loaded")
sendMessage(input)
log("Input received")
generateResponse(input)
log("Generating response")
response
log("Response sent")
sendResponse(response)
endChat()
log("Chat ended")

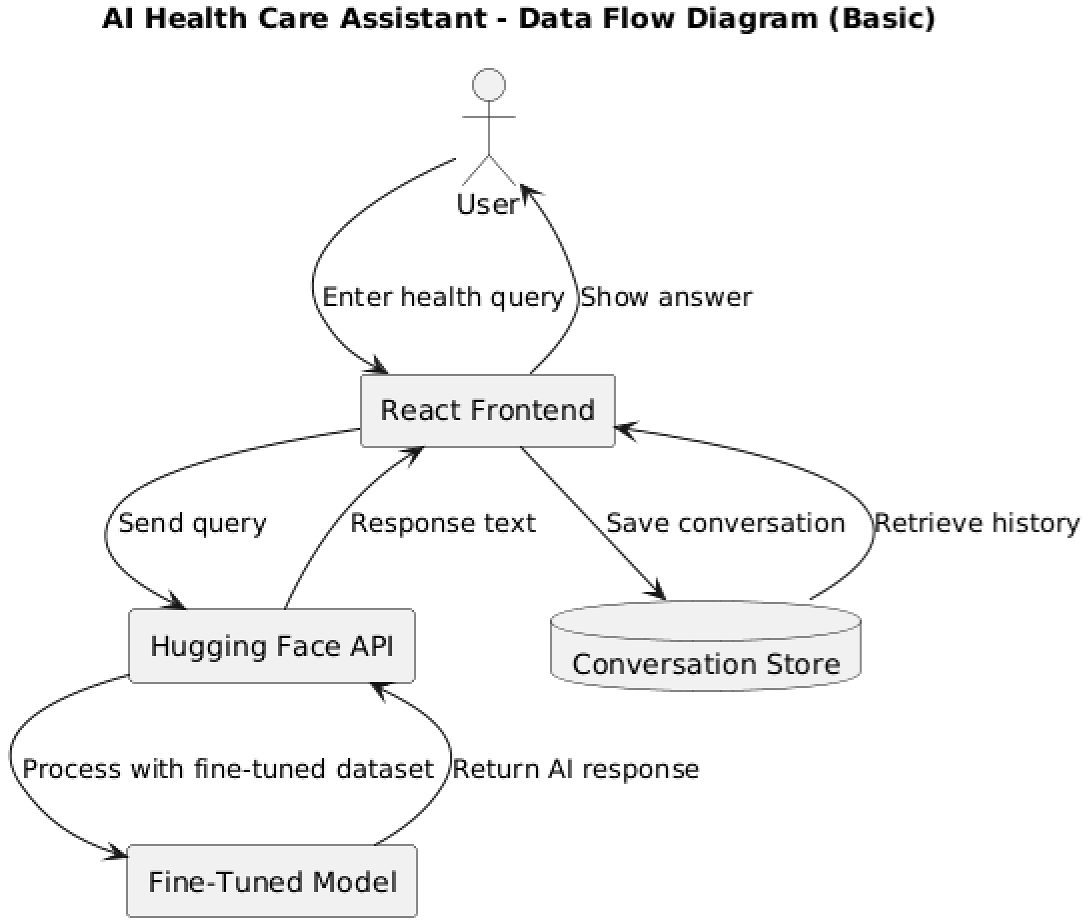
Class Diagram:



Activity Diagram:



Data Flow Diagram:



Thank You…

- 1. F. Anil Kumar – 22N31A7313
- 2. R. Varaprasad – 22N31A7356
- 3. S. Srujana – 23N35A7306